Chapter 20: Treatment-Confounder Feedback

Why Traditional Regression Fails for Longitudinal Telco Retention Programs & How G-Methods Resolve It

1. The Time-Varying Confounding Dilemma in Customer Retention

In multi-period customer retention programs, subscriber treatment status is updated dynamically over time. For example, a telecom operator evaluates customers monthly at k = 0, 1, ..., K, offering retention discounts or service upgrades ($A_k = 1$) based on ongoing service usage, bill amounts, and network complaint logs (L_k). This dynamic feedback loop creates a fundamental analytical challenge known as **Treatment-Confounder Feedback**.

As formalized in Chapter 20 of Hernán & Robins (What If), a time-varying covariate L_k plays a dual role: it acts simultaneously as a **confounder** for future treatment A_k and as an **intermediate variable (mediator)** affected by prior treatment A_{k-1}. Standard regression models and propensity matching cannot handle variables with this dual identity.

2. Structural DAG: The Feedback Loop & Unobserved Confounding

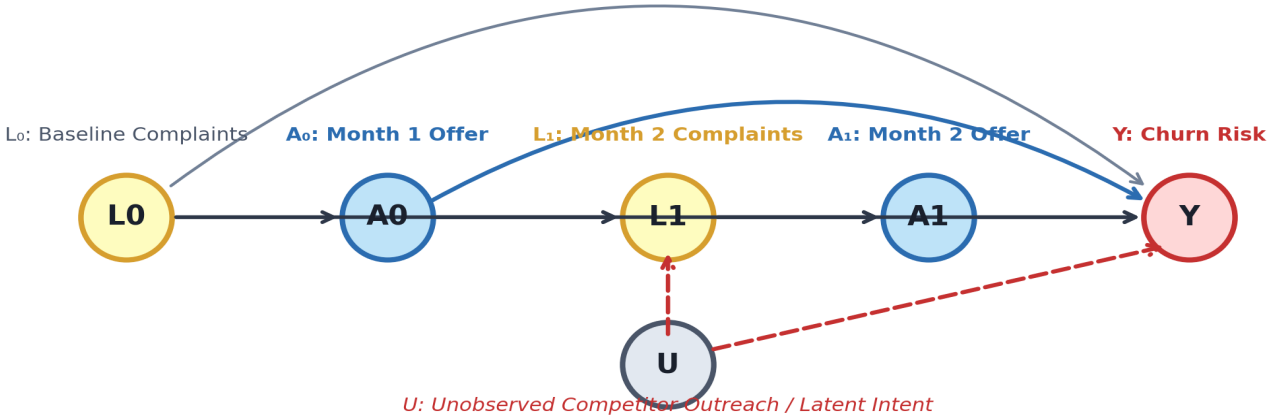


Figure 20.1: Causal DAG for a 2-period retention intervention. Month 1 offer A_0 affects Month 2 complaints L_1, which determines Month 2 offer A_1 and Churn Y. Unobserved competitor outreach U affects both L_1 and Y.

3. The Catch-22: Why Standard Regression Is Fundamentally Flawed

Suppose an analyst fits a standard multi-period logistic or Cox regression model for churn Y adjusting for treatment history (A_0, A_1) and covariate history (L_0, L_1). This standard approach fails regardless of whether L_1 is included or excluded in the regression equation:

> **FLAW #1: OVERADJUSTMENT & PATHWAY BLOCKING**
>
> **Flaw 1: Adjusting for L_1 blocks the causal effect of A_0 (Overadjustment Bias).**
> Part of the true benefit of Month 1 offer ($A_0 = 1$) is reducing complaint calls in Month 2 ($L_1 = 0$). Holding L_1 constant in a standard regression blocks this indirect causal pathway ($A_0 \rightarrow L_1 \rightarrow Y$), thereby underestimating the total effectiveness of the initial retention offer.

FLAW #2: COLLIDER STRATIFICATION BIAS

Flaw 2: Adjusting for L_1 induces Selection Bias / Collider Stratification Bias.
Unobserved factors U (e.g. competitor pricing or unrecorded network outages) affect both complaints (L_1) and churn (Y). In the DAG, L_1 is a **collider** on the path $A_0 \rightarrow L_1 \leftarrow U \rightarrow Y$. Conditioning on L_1 in regression opens a non-causal backdoor path, creating spurious correlation between A_0 and churn Y.

Conversely, if the analyst **does not adjust for L_1**, L_1 remains an unblocked confounder for Month 2 treatment A_1 and churn Y. Thus, **standard regression faces an inescapable Catch-22**: adjusting for L_1 biases the estimate of A_0, while omitting L_1 biases the estimate of A_1.

4. Empirical Case Study: Simulating Telco Retention Pipeline Bias

To quantify this bias in production telecom settings, consider a simulated cohort of 100,000 subscribers tracked over 2 months. True data generating process: Month 1 offer (A_0) reduces Month 2 complaints (L_1) by 30%. True 2-period treatment effect of continuous retention (A_0=1, A_1=1 vs A_0=0, A_1=0) is a **-6.0 percentage point reduction in churn risk**.

Model / Estimator	Adjustment for L_1	Estimated ATE (A_0=1, A_1=1)	Bias vs. True ATE (-6.0%)	Methodological Flaw Identified
Naive Unadjusted Contrast	None	**-11.4%**	+5.4% (Severe Overestimation)	Confounding by L_0 and L_1 unaddressed.
Standard Multivariable Logistic	Adjusts for L_0 and L_1	**-1.8%**	-4.2% (Severe Underestimation)	Overadjustment bias (blocks $A_0 \rightarrow L_1 \rightarrow Y$) + Collider bias via U.
Standard Cox Proportional Hazards	Time-Varying $L_1(t)$	**-2.2%** (HR = 0.81)	-3.8% (Biased HR)	Conditioning on time-varying $L_1(t)$ causes collider stratification bias.
Standard Propensity Score Matching	Matched on L_0 and L_1	**-2.0%**	-4.0% (Biased ATT)	Matching on intermediate L_1 removes genuine treatment effect.
IP Weighted MSM (G-Method)	Weighted by SW^A_k	**-5.9%**	**-0.1% (Unbiased)**	Removes $L_k \rightarrow A_k$ arrows without conditioning on L_1.
Longitudinal G-Formula	Integrates over L_1	**-6.1%**	**+0.1% (Unbiased)**	Simulates counterfactual L_1 trajectories under intervention.

5. How G-Methods Resolve Treatment-Confounder Feedback

To obtain unbiased estimates of time-varying treatment strategies in the presence of treatment-confounder feedback, epidemiologists and statisticians developed **G-Methods** (Robins, 1986; Hernán & Robins, 2024). There are three primary G-methods, each taking a distinct approach to bypass the Catch-22:

THE THREE G-METHODS FOR LONGITUDINAL DATA

1. Inverse Probability Weighting (IPW) of Marginal Structural Models (MSMs):
IPW reweights the longitudinal population using stabilized weights $SW^{A}_{k} = \prod_{t=0..k} [f(A_t \mid \bar{A}_{t-1}) / f(A_t \mid \bar{A}_{t-1}, \bar{L}_t)]$. In the weighted pseudo-population, the arrows $L_k \rightarrow A_k$ are mathematically severed. Confounding is eliminated without ever conditioning on L_1 in the outcome model, preserving all indirect pathways $A_0 \rightarrow L_1 \rightarrow Y$ and avoiding collider bias.

2. Parametric G-Formula (Standardization):
The G-formula models both the outcome Y given history $(\bar{A}_k, \bar{L}_k)$ and time-varying covariates L_k given past history. It algorithmically simulates step-by-step counterfactual evolution of L_1, L_2, ... under strategy g.

3. G-Estimation of Structural Nested Models (SNMs):
G-estimation models average causal contrasts directly via unblipping functions, testing for sequential exchangeability.

6. Production Architecture for Industrial Telecom Pipelines

When engineering longitudinal churn prediction and campaign evaluation systems in enterprise environments, data science teams should implement the following decision rules:

- **Audit Data for Feedback Loops:** Check if past retention campaigns (A_{t-1}) alter current customer state features (L_t). If $P(L_t \mid A_{t-1}, L_{t-1}) \neq P(L_t \mid L_{t-1})$, treatment-confounder feedback is present.
- **Deprecate Standard Regression for Policy Evaluation:** Do not use standard logistic regression or Cox proportional hazards with time-varying covariates L(t) to measure the causal impact of multi-period promotional strategies.
- **Implement IPW MSMs for Static Regimes:** For evaluating fixed multi-month retention policies (e.g. 6-month promotional cycles), fit a Marginal Structural Model weighted by stabilized longitudinal weights SW^{A}_{k}.
- **Implement G-Formula / TMLE for Dynamic Regimes:** For evaluating personalized dynamic rules (e.g. 'offer 20% discount whenever monthly usage drops by >15%'), use the Parametric G-Formula or Longitudinal Targeted Minimum Loss-Based Estimation (L-TMLE).